# MSG No. 15.86 $C2/c1'$ [ Type II, monoclinic ]

Table 1: Wyckoff site: `4a`, site symmetry: `-11'`

| No. | position | mapping |
|---|---|---|
| 1 | $[0,\ 0,\ 0]$ | `[1,3,9,11]` |
| 2 | $\left[0,\ 0,\ \frac{1}{2}\right]$ | `[2,4,10,12]` |
| 3 | $\left[\frac{1}{2},\ \frac{1}{2},\ 0\right]$ | `[5,7,13,15]` |
| 4 | $\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$ | `[6,8,14,16]` |

Table 2: Wyckoff site: `4b`, site symmetry: `-11'`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[0,\ \frac{1}{2},\ 0\right]$ | `[1,3,9,11]` |
| 2 | $\left[0,\ \frac{1}{2},\ \frac{1}{2}\right]$ | `[2,4,10,12]` |
| 3 | $\left[\frac{1}{2},\ 0,\ 0\right]$ | `[5,7,13,15]` |
| 4 | $\left[\frac{1}{2},\ 0,\ \frac{1}{2}\right]$ | `[6,8,14,16]` |

Table 3: Wyckoff site: `4c`, site symmetry: `-11'`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{1}{4},\ \frac{1}{4},\ 0\right]$ | `[1,7,9,15]` |
| 2 | $\left[\frac{3}{4},\ \frac{1}{4},\ \frac{1}{2}\right]$ | `[2,8,10,16]` |
| 3 | $\left[\frac{3}{4},\ \frac{3}{4},\ 0\right]$ | `[3,5,11,13]` |
| 4 | $\left[\frac{1}{4},\ \frac{3}{4},\ \frac{1}{2}\right]$ | `[4,6,12,14]` |

Table 4: Wyckoff site: `4d`, site symmetry: `-11'`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{1}{4},\ \frac{1}{4},\ \frac{1}{2}\right]$ | `[1,7,9,15]` |
| 2 | $\left[\frac{3}{4},\ \frac{1}{4},\ 0\right]$ | `[2,8,10,16]` |
| 3 | $\left[\frac{3}{4},\ \frac{3}{4},\ \frac{1}{2}\right]$ | `[3,5,11,13]` |
| 4 | $\left[\frac{1}{4},\ \frac{3}{4},\ 0\right]$ | `[4,6,12,14]` |

Table 5: Wyckoff site: `4e`, site symmetry: `21'`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[0,\ y,\ \frac{1}{4}\right]$ | `[1,2,9,10]` |
| 2 | $\left[0,\ -y,\ \frac{3}{4}\right]$ | `[3,4,11,12]` |
| 3 | $\left[\frac{1}{2},\ y+\frac{1}{2},\ \frac{1}{4}\right]$ | `[5,6,13,14]` |
| 4 | $\left[\frac{1}{2},\ \frac{1}{2}-y,\ \frac{3}{4}\right]$ | `[7,8,15,16]` |

Table 6: Wyckoff site: `8f`, site symmetry: `11'`

| No. | position | mapping |
|---|---|---|
| 1 | $[x,\, y,\, z]$ | `[1,9]` |
| 2 | $\left[-x,\, y,\, \frac{1}{2}-z\right]$ | `[2,10]` |
| 3 | $[-x,\, -y,\, -z]$ | `[3,11]` |
| 4 | $\left[x,\, -y,\, z+\frac{1}{2}\right]$ | `[4,12]` |
| 5 | $\left[x+\frac{1}{2},\, y+\frac{1}{2},\, z\right]$ | `[5,13]` |
| 6 | $\left[\frac{1}{2}-x,\, y+\frac{1}{2},\, \frac{1}{2}-z\right]$ | `[6,14]` |
| 7 | $\left[\frac{1}{2}-x,\, \frac{1}{2}-y,\, -z\right]$ | `[7,15]` |
| 8 | $\left[x+\frac{1}{2},\, \frac{1}{2}-y,\, z+\frac{1}{2}\right]$ | `[8,16]` |